

1. interpret information about how solid materials can differ with respect to properties such as:
  - melting points;
  - strength (in tension or compression);
  - stiffness;
  - hardness;
  - density;
2. relate properties to the uses of materials such as plastics, rubbers and fibres;
3. relate the effectiveness and durability of a product to the materials used to make it;
4. interpret information about the properties of materials such as plastics, rubbers and fibres to assess the suitability of these materials for particular purposes.
5. with respect to data from the measurement of properties of materials:
  - uses data rather than opinion in justifying an explanation;
  - can suggest reasons why a measurement may be inaccurate;
  - can suggest reasons why several measurements of the same quantity may give different results;
  - when asked to evaluate data, makes reference to its reliability (i.e. is it repeatable?);
  - can calculate the mean of a set of repeated measurements;
  - from a set of repeated measurements of a quantity, uses the mean as the best estimate of the true value;
  - can explain why repeating measurements leads to a better estimate of the quantity;
  - can make a sensible suggestion about the range within which the true value of a measured quantity probably lies;
  - **can justify the claim that there is/is not a 'real difference' between two measurements of the same quantity;**
  - can identify any outliers in a set of data, and give reasons for including or discarding them;
  - can identify, in a plan for an investigation of the effect of a factor on an outcome, the fact that other factors are controlled as a positive feature, or the fact that they are not as a design flaw;
  - **can explain why it is necessary to control all factors thought likely to affect the outcome other than the one being investigated.**